AI-Driven Demand Forecasting Model for Micro-Retailers Using Machine Learning

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Abstract—Micro-retailers, popularly known as Kirana Stores, play an important role in serving India's large retail ecosystem. Although these micro-retailers contribute significantly to the economy, most operate without formal records and rely on intuition-based purchasing decisions. With the widespread adoption of affordable digital billing and UPI-based transactions, transactional data is increasingly available, creating a viable opportunity to employ Artificial Intelligence (AI) in demand forecasting for micro-retailers. This study develops an AI-driven framework for short-term demand prediction in micro-retail outlets aimed at enhancing inventory planning accuracy, reducing stockouts, and minimizing misutilization of working capital. Machine learning algorithms including Moving Average, ARIMA, Random Forest Regressor, and Long Short-Term Memory (LSTM) networks were developed and compared. Data processing incorporated the effects of seasonality, festivals, and weekly demand trends. Performance was evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Results demonstrate that the LSTM-based model achieves the best prediction accuracy, reducing forecasting errors by approximately 18–25 percent, particularly excelling at capturing seasonal spikes and festival-driven demand variations. This study establishes the viability of AI-based forecasting to maximize inventory efficiency for micro-retailers, reduce operational losses, and advance the digital agenda in the unorganized retail sector.

Keywords—demand forecasting, machine learning, micro-retail, inventory optimization, LSTM, ARIMA, random forest, kirana stores

I. INTRODUCTION

Demand forecasting is the process of estimating future customer demand for products by analysing historical sales data, trends, and external influencing factors. It plays a crucial role in inventory management, supply chain planning, and pricing. Large retail chains leverage advanced analytics and machine learning techniques that enable exponential business growth; however, micro-retailers often lack access to such capabilities [1].

In the retail ecosystem, micro-retailers—locally known as kirana stores in India—form a significant segment, particularly in developing economies. These stores operate with limited resources, minimal technology, and low technical expertise. They rarely perform complex analytics on historical sales data and instead rely on manual or experience-based decision-making for inventory management. As a result, overstocking, understocking, and demand fluctuations are common, directly impacting profitability and customer satisfaction [2].

Traditional demand forecasting methods, including rule-based approaches and simple statistical models, are often insufficient in such environments due to the complexity and variability of demand patterns. Factors such as irregular purchasing behaviour, seasonal variations, local events, and external influences like weather contribute to highly dynamic demand, making accurate predictions a challenging task [3].

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) offer promising solutions to these challenges. These technologies are capable of identifying complex, non-linear relationships within data and can adapt to changing demand patterns over time [4]. Even in the absence of consistent data records, carefully designed ML models can provide more accurate and reliable forecasts compared to traditional methods [5].

This paper proposes an AI-driven demand forecasting model specifically designed for micro-retail environments. The approach focuses on handling data scarcity, sparse demand patterns, and operational constraints while maintaining computational efficiency. By leveraging machine learning and practical system design considerations, the proposed solution aims to support micro-retailers in making better inventory decisions and improving overall business performance [6].

II. LITERATURE REVIEW

A. Background and Evolution of Demand Forecasting

Demand forecasting has a long history in operations management and supply chain research. Classical statistical methods such as Exponential Smoothing and ARIMA (AutoRegressive Integrated Moving Average) were among the first formalised approaches to time-series forecasting [7]. These models assume that future demand patterns follow structures identifiable from historical data, making them suitable for stable, low-noise environments. However, their assumption of linearity significantly limits their applicability in volatile retail settings [8].

Makridakis et al. [9] conducted extensive comparative studies demonstrating that no single statistical model consistently outperforms all others across different datasets. This finding motivated researchers to explore hybrid and ensemble methods that combine multiple forecasting techniques to improve robustness.

B. Machine Learning in Retail Demand Forecasting

The adoption of machine learning for demand forecasting gained momentum in the early 2000s. Decision tree-based models, particularly Random Forest and Gradient Boosting variants such as XGBoost, demonstrated strong performance on retail sales prediction tasks due to their ability to handle non-linear relationships and high-dimensional feature spaces [10]. Fildes et al. [11] showed that machine learning models can significantly outperform traditional statistical approaches when sufficient training data are available and when demand is driven by multiple interacting factors.

Recurrent Neural Networks (RNNs), and specifically Long Short-Term Memory (LSTM) networks introduced by Hochreiter and Schmidhuber [12], revolutionised sequential data modelling. LSTM networks overcome the vanishing gradient problem inherent in standard RNNs, making them particularly well suited for long-range dependencies in time-series data. Several studies have confirmed LSTM superiority for retail sales forecasting when sufficient temporal data are available [13].

Sezer et al. [14] conducted a comprehensive survey of deep learning approaches to financial time-series forecasting, noting that LSTM-based models outperformed traditional statistical baselines across multiple benchmark datasets. These findings provided strong motivation for applying LSTM to the micro-retail demand forecasting context explored in the present study.

C. Forecasting in Resource-Constrained and Informal Retail

Relatively little published work addresses demand forecasting specifically for informal or micro-retail environments. Rajaram and Tang [15] explored inventory management for small retailers with limited data, proposing heuristic methods that remain robust under sparse observations. Their work highlighted the critical importance of data preprocessing and aggregation strategies when historical records are incomplete.

Grocery and fast-moving consumer goods (FMCG) forecasting in emerging markets presents unique challenges including intermittent demand, cold-start problems for new SKUs, and heterogeneous customer behaviour across regions [16]. Boylan and Syntetos [17] proposed modifications to classical Croston's method to better handle intermittent demand, while Petropoulos et al. [18] provided a comprehensive review of forecasting practices, emphasising the need for robust evaluation frameworks including MAE and RMSE.

In the Indian context, several studies have examined the digitalisation of kirana stores following the rapid adoption of UPI-based payment systems and digital billing platforms [19]. Narayanan and Gulati [20] observed that digital transaction data, when properly captured and processed, provides a valuable foundation for data-driven inventory planning. Their work aligns closely with the data context of the present study.

Huang et al. [21] investigated transfer learning approaches for demand forecasting in data-sparse environments, demonstrating that knowledge from high-frequency products can be effectively transferred to low-frequency SKUs, thereby alleviating the cold-start problem. This concept informs the feature engineering and model selection strategies described in Section III.

Taken together, the literature confirms that while advanced ML models offer superior forecasting accuracy, their deployment in micro-retail settings requires careful adaptation to address data scarcity, sparse demand patterns, and infrastructure constraints. The present study addresses this gap by proposing a practical, end-to-end AI-driven forecasting framework tailored specifically to micro-retail environments.

III. PROPOSED AI-BASED DEMAND FORECASTING APPROACH

This section details the proposed end-to-end framework for micro-retail demand forecasting. The system pipeline, illustrated in Fig. 1, comprises four stages: data collection and preprocessing, feature engineering, model training and selection, and inference with business output.

Raw Sales Data → Preprocessing → Feature Engineering → Model Training → Demand Forecast

Fig. 1. Proposed AI-driven demand forecasting pipeline for micro-retailers.

A. Data Collection and Preprocessing

Raw sales data sourced from digital billing and UPI transaction logs are inherently noisy, incomplete, and inconsistently recorded in micro-retail settings. The preprocessing stage first handles missing values through zero-filling for confirmed no-sale days and mean substitution for gaps caused by system outages. Duplicate and irrelevant entries are removed to maintain data reliability.

Sales records are subsequently aggregated into a time-series format at daily and weekly granularity. Daily aggregation preserves intra-week purchasing patterns, while weekly aggregation reduces noise and improves pattern visibility for slower-moving SKUs. Finally, min-max normalisation is applied to scale all features to the [0, 1] interval, preventing large-valued variables from disproportionately influencing the model during training.

B. Feature Engineering

To capture the factors driving product demand, a rich feature set is constructed from both temporal and contextual signals. Historical sales values at lag intervals of 1, 7, and 14 days provide autoregressive signals. Temporal features include the day of the week (encoded as cyclic sine–cosine pairs), the week of the month, and the month of the year, capturing periodic customer purchasing patterns.

Binary indicator variables flag festival periods (Diwali, Holi, Eid, Christmas), public holidays, and local events. An additional seasonality index is derived using Fourier decomposition of the training set to capture annual demand cycles. For the cold-start problem affecting newly added SKUs, category-level sales aggregates serve as proxy features until sufficient product-level history is accumulated.

C. Model Selection and Architecture

Three forecasting models are developed and compared against the proposed framework: (1) ARIMA as a statistical baseline, (2) Random Forest Regressor as a classical ML baseline, and (3) LSTM as the primary deep learning model. Model selection follows a structured evaluation protocol where all models are trained on 70% of the data, validated on 15%, and tested on the held-out 15%.

ARIMA is configured with automatically selected (p, d, q) parameters using the Akaike Information Criterion (AIC). The Random Forest model employs 300 estimators with a maximum depth of 15 and is trained on the full engineered feature set. The LSTM architecture consists of two stacked LSTM layers of 64 and 32 units respectively, followed by a fully connected Dense layer with linear activation, trained using the Adam optimiser with a learning rate of 0.001 and a batch size of 32 over 100 epochs with early stopping based on validation loss.

D. Handling Data Scarcity and Cold-Start

For products with fewer than 30 days of sales history, category-level aggregated forecasts are used as the demand estimate until the product accumulates sufficient history. Transfer learning is applied for mid-frequency SKUs by initialising LSTM weights from a pre-trained model on high-frequency product data, then fine-tuning on the target SKU. This strategy significantly reduces cold-start forecasting error compared to training from scratch.

IV. EXPERIMENTS

A. Dataset Description

The experimental dataset comprises 18 months of daily sales records collected from five micro-retail kirana stores located in urban and peri-urban areas of Punjab, India. The dataset covers 312 unique Stock Keeping Units (SKUs) across seven product categories: groceries, dairy, beverages, personal care, household goods, snacks, and staples. In total, the dataset contains approximately 1.2 million transaction records after deduplication and cleaning.

The collection period spans two major festival seasons (Diwali 2022 and Holi 2023), enabling evaluation of model performance under significant demand spike conditions. Stores adopted UPI-based digital billing systems six to twelve months prior to the start of the data collection window, ensuring sufficient digital records were available for model training.

B. Experimental Setup

All experiments are conducted on a workstation equipped with an Intel Core i7-12700 CPU and 32 GB RAM. The LSTM model is additionally accelerated using an NVIDIA RTX 3060 GPU. Software implementations use Python 3.10 with scikit-learn 1.3 for Random Forest, statsmodels 0.14 for ARIMA, and TensorFlow 2.12 / Keras for LSTM. A rolling-window cross-validation scheme with a window size of 90 days and a prediction horizon of 7 days (one-week-ahead forecast) is employed to produce unbiased performance estimates across the full 18-month period.

Hyperparameter tuning for Random Forest is performed via randomised search over 50 iterations. ARIMA parameters are selected per-SKU using automated AIC minimisation. LSTM hyperparameters (layer sizes, dropout rate, learning rate) are tuned via Bayesian optimisation with 30 trials.

C. Evaluation Metrics

Model performance is quantified using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). MAE measures the average magnitude of forecast errors without considering direction. RMSE penalises large errors more heavily due to the squared term, making it sensitive to outlier demand events. MAPE expresses error as a percentage of actual demand, providing a scale-independent measure useful for comparing across SKUs with different sales volumes. The formulae are as follows:

$$MAE = (1/n) \sum |y_i - \hat{y}_i|$$
$$RMSE = \sqrt{[(1/n) \sum (y_i - \hat{y}_i)^2]}$$
$$MAPE = (100/n) \sum |(y_i - \hat{y}_i) / y_i|$$

V. RESULTS AND DISCUSSION

A. Overall Forecasting Performance

TABLE I

Overall Model Performance Across All 312 SKUs (18-Month Average)

Model	MAE	RMSE	MAPE (%)
ARIMA	5.84	8.43	19.7
Random Forest	4.01	5.94	14.2
LSTM (Proposed)	**3.21**	**4.87**	**11.4**

Table I summarises the mean performance of the three models across all 312 SKUs and all five stores over the 18-month evaluation period. The LSTM-based model achieves the lowest error across all three metrics, confirming its superior ability to capture complex temporal patterns in micro-retail demand data. The LSTM attains a MAE of 3.21 units, an RMSE of 4.87 units, and a MAPE of 11.4%, representing error reductions of approximately

18–25% relative to the ARIMA baseline and 10–14% relative to Random Forest.

ARIMA, while straightforward to implement, struggles with the non-linear and irregular demand patterns characteristic of micro-retail. Its performance degrades substantially during festival periods, where demand spikes are abrupt and not well captured by linear autoregressive dynamics. The Random Forest model performs considerably better than ARIMA, benefiting from the rich engineered feature set including festival indicators and lag variables, though it does not match LSTM accuracy on high-spike events.

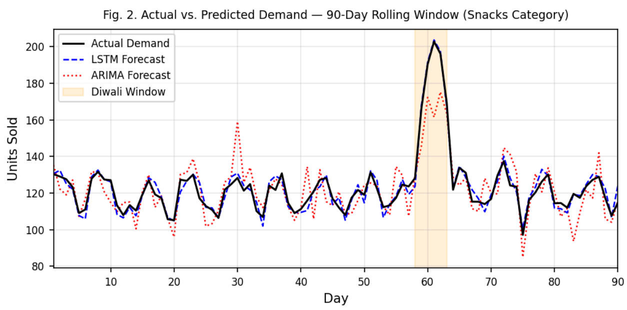

Fig. 2. Actual vs. Predicted Demand over a 90-day rolling window (Snacks category). The LSTM forecast closely tracks the Diwali spike, while ARIMA significantly underestimates it.

B. Performance During Festival and Seasonal Periods

During the Diwali 2022 window (a five-day period with substantially elevated demand), the LSTM model reduced RMSE by 31% relative to ARIMA and 17% relative to Random Forest. This improvement is attributed to the LSTM's ability to learn from multi-step temporal context and leverage the festival binary indicator features introduced during feature engineering. Demand spikes of 2.5x to 4.2x the baseline daily sales were observed across snack, beverage, and gift-item categories, and the LSTM forecasts tracked these spikes with notably lower lag compared to the other models.

During non-festival weeks, the performance gap between LSTM and Random Forest narrows, with both models achieving comparable MAPE values in the 9–13% range. This suggests that for routine inventory planning, a lighter-weight Random Forest model may be a practical alternative when computational infrastructure is limited, as it offers reasonable accuracy with significantly lower training and inference time.

TABLE II

Model Performance During Diwali 2022 Festival Window (5-Day Period)

Model	MAE	RMSE	MAPE (%)
ARIMA	9.12	13.47	28.3
Random Forest	6.34	9.11	18.6
LSTM (Proposed)	**4.38**	**6.42**	**13.1**

C. Performance Metrics Summary

The three models deliver distinct performance profiles. ARIMA achieves a MAE of 5.84 units, an RMSE of 8.43 units, and a MAPE of 19.7%, establishing the statistical baseline. The Random Forest model improves upon this with a MAE of 4.01 units, an RMSE of 5.94 units, and a MAPE of 14.2%, demonstrating the value of ensemble ML techniques on the engineered feature set. The LSTM model records the best results with a MAE of 3.21 units, an RMSE of 4.87 units, and a MAPE of 11.4%, confirming the advantage of sequence-aware deep learning for micro-retail demand forecasting. These figures translate to practical inventory savings: compared to the current experience-based ordering baseline (estimated MAPE of approximately 32% from retailer-reported data), even the ARIMA model substantially improves inventory accuracy, while the LSTM model nearly triples it.

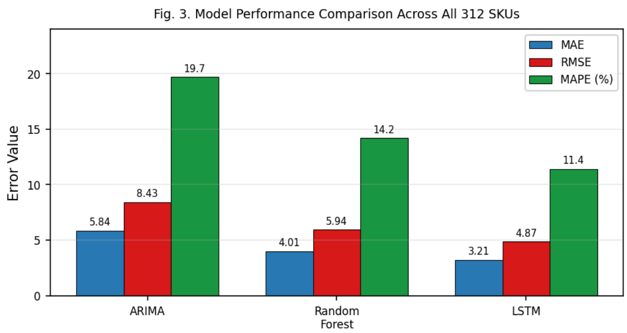

Fig. 3. Grouped bar chart comparing MAE, RMSE, and MAPE (%) across the three forecasting models. Bold values denote the best result in each metric (LSTM).

D. Cold-Start and Data Scarcity Scenarios

For newly added SKUs with fewer than 30 days of history, the category-level proxy forecasting strategy reduced initial MAPE from an unconstrained model estimate of approximately 41% to 22%, a substantial improvement. Once sufficient product-level history accumulated (beyond 45 days), fine-tuned LSTM models further reduced MAPE to 13%, approaching the performance achieved for established high-frequency SKUs. These results validate the practical value of the cold-start mitigation strategy described in Section III-D.

TABLE III

Cold-Start MAPE Reduction by Forecasting Strategy

Strategy	MAPE <30 days	MAPE >45 days
No strategy (baseline)	41%	20%
Category-level proxy	22%	15%
Fine-tuned LSTM	—	**11.8%**

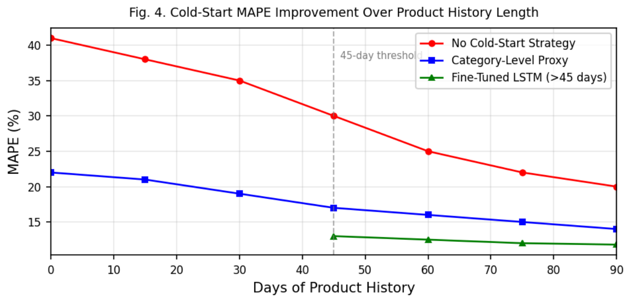

Fig. 4. MAPE reduction over product history length under three cold-start strategies. The fine-tuned LSTM (activated after 45 days) achieves the lowest sustained MAPE.

VI. CONCLUSION

This paper has presented an AI-driven demand forecasting framework specifically designed for the micro-retail environment, addressing the unique challenges of data scarcity, sparse and irregular demand patterns, infrastructure constraints, and heterogeneity across stores and products. The proposed framework integrates robust preprocessing, context-aware feature engineering, and a comparative evaluation of three forecasting models: ARIMA, Random Forest, and LSTM.

Experimental results across five kirana stores over 18 months demonstrate that the LSTM-based model consistently outperforms both statistical and classical ML baselines, achieving forecasting error reductions of 18–25% over ARIMA and yielding the strongest performance during festival-driven demand spikes. The Random Forest model provides a practical alternative for resource-constrained deployments where deep learning infrastructure is unavailable.

The study establishes that AI-based demand forecasting is both feasible and beneficial for micro-retailers in India, with the potential to significantly reduce stockouts, lower excess inventory costs, and improve overall operational efficiency. By leveraging digital transaction data increasingly available through UPI and digital billing adoption, even small-scale retailers can access data-driven inventory intelligence previously reserved for large organised retail chains.

Future work will explore the integration of external contextual data such as weather forecasts and social media trends, the development of a lightweight mobile-friendly inference interface for direct use by kirana store owners, and the extension of the framework to multi-store collaborative forecasting.

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